

Casimir force between doped silicon slabs

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Abstract

We evaluate the Casimir force for the particular case of silicon material including mirror thickness impact and finite conductivity influence. We show a new interesting behavior related to the slab thickness. We compare the results for intrinsic and doped silicon with traditional metals such as gold.

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1. Introduction

The “Casimir force” is a macroscopic physical manifestation of the zero point energy. This force whose existence was theoretically predicted in 1948 by the Dutch physicist Casimir [1] had been precisely measured only in 1997 by the American team of Lamoreaux [2]. Between two uncharged infinite parallel plates, with perfect conductivity, separated by a distance d , this force per unit area has the magnitude:

$$F_c = \frac{\pi^2 \hbar c}{240 d^4}. \quad (1)$$

This electrodynamics effect is actively studied on macro or microscopic structures to predict the Casimir force with high precision [3–7].

The historical strategy to evaluate the vacuum fluctuation effects is based on the potential energy calculation integrating over all cavity eigen modes [8,9]. The second computation technique consists in solving the coupled equations of atom motions and quantum electromagnetic field. The third technique had been developed by Lifshitz [10] for two dielectric plates separated by a vacuum gap. This approach uses the concept of optical transfer functions of slabs in glance to define the electromagnetic radiation pressure in and outside the cavity.

In this Letter, we follow this last methodology basing on Reynaud published works [11] to give the general expression of the Casimir effect for any slab thickness and materials, perfectly suitable for any nanomechanical structures (NEMS) modeling.

We show the influence of the slab thickness on the corrective factor of the Casimir force for real silicon slabs at different doping levels. We finally compare silicon with the typical case of gold.

We do not include the temperature effect, which will remain weak for distances shorter than the thermal wavelength equal to 7 μm at ambient temperature. This value is largely higher than the dimensions considered in nanostructures [12]. We do not consider in our computations roughness effect [13] that is out of field of this Letter.

2. Theory review

We consider the simple geometry depicted in Fig. 1. The slabs are made of the same material, and the media between slabs is the vacuum. This structure is also the basic one we typically find in NEMS architectures.

We have to correct the ideal Casimir force (1) including the imperfect conductivity of the materials and the finite thickness of the slabs. Classically, the corrective factor to apply to the force is defined as following:

$$\eta = \frac{\langle \hat{P} \rangle}{F_c}, \quad (2)$$

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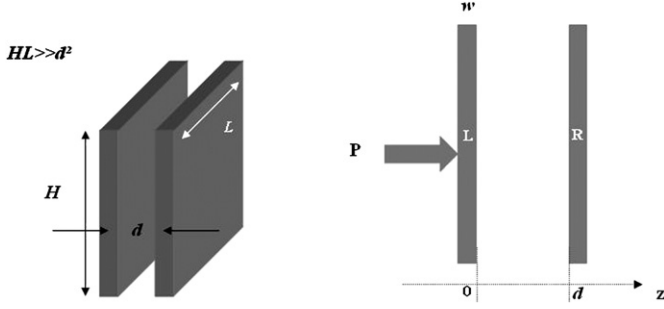


Fig. 1. Two slabs in glance—3D view and notation definition.

F_c is the force per unit area shown in Eq. (1) and $\mathbf{P}$ is the resulting radiation pressure on the cavity. As already demonstrated by Reynaud [11], we express $\mathbf{P}$:

$$\langle \hat{P} \rangle = \sum_m \iint \frac{dk_x dk_y}{4\pi^2} \int \frac{dk_z}{2\pi} \cos^2(\theta_m(k_x, k_y, k_z)) \times \hbar \omega_m(k_x, k_y, k_z) (1 - A_m(k_x, k_y, k_z)), \quad (3)$$

(k_x, k_y, k_z) are the wave vector components of the incident electromagnetic field in the vacuum. The symbol m represents s and p polarizations. $\theta_m(k_x, k_y, k_z)$ is the incidence angle of the input field. $\omega_m(k_x, k_y, k_z)$ is the angular frequency depending on the wave vector through the dispersion relationship. $A_m(k_x, k_y, k_z)$ is the Airy function of the cavity that mainly depends on the slab reflection coefficients.

This function can be written as:

$$A_m(k_x, k_y, k_z) = \frac{1 - |r_{mslab}(k_x, k_y, k_z)|^2}{|1 - r_{mslab}(k_x, k_y, k_z)^2 \exp(2ik_z d)|^2}, \quad (4)$$

$r_{mslab}(\omega, k_z)$ is the reflection coefficient of one slab for s and p polarizations, depending simultaneously on the thickness and dielectric function $\varepsilon(\omega)$ of material.

The expression (4) can be broken up in two parts:

$$(1 - A_m(\omega, k_z)) \equiv f_m(\omega, k_z) + f_m^*(\omega, k_z),$$

$$f_m(\omega, k_z) = \frac{r_{mslab}^2(\omega, k_z) \exp(2ik_z d)}{1 - r_{mslab}^2(\omega, k_z) \exp(2ik_z d)}. \quad (5)$$

We set the following dimensionless variables:

$$k_z = i\kappa, \quad K = \kappa d,$$

$$\omega = i\xi, \quad \Omega = \frac{\xi d}{c},$$

$$x = \frac{w}{d} \sqrt{K^2 + \Omega^2(\varepsilon - 1)},$$

$$\varepsilon = \varepsilon(\omega) = \varepsilon\left(i \frac{c\Omega}{d}\right). \quad (6)$$

From the previous equations (6) and after some mathematical computations, we finally get a general expression of the corrective factor explicitly given for any width:

$$\eta = \frac{120}{\pi^4} \sum_m \int_0^{+\infty} dK K^3 \int_0^K \frac{d\Omega}{K} f(r_m^2, K, x),$$

$$f(r_m^2, K, x) = \frac{r_m^2 (1 - e^{-2x})^2 e^{-2K}}{(1 - r_m^2 e^{-2x})^2 - r_m^2 (1 - e^{-2x})^2 e^{-2K}},$$

$$r_p = \frac{\sqrt{K^2 + \Omega^2(\varepsilon - 1)} - \varepsilon K}{\sqrt{K^2 + \Omega^2(\varepsilon - 1)} + \varepsilon K},$$

$$r_s = \frac{\sqrt{K^2 + \Omega^2(\varepsilon - 1)} - K}{\sqrt{K^2 + \Omega^2(\varepsilon - 1)} + K}. \quad (7)$$

The corrective factor depends on K , x and reflection coefficients r_s and r_p at each interface for s and p polarization respectively. This general expression includes the conductivity of the material for any slab thickness and can be easily evaluated by numerical way (trapezoid method for example). The material can be any metals or silicon with different doping levels. We use this expression for all further computations.

3. Application to silicon

We now compute the corrective factor for silicon materials using (7) with the methodology described by Lambrecht and Reynaud in [14]. In this way, we build a simple permittivity model $\varepsilon(\omega)$ (8), based on the Drude expression to include the charge carrier effect and on the Edwards work [15] to define the intrinsic part $\varepsilon_{intrinsic}(\omega)$.

$$\varepsilon(\omega) = \varepsilon_{intrinsic}(\omega) - \frac{\omega_p^2}{\omega(\omega - i\gamma)},$$

$$\omega_p = \sqrt{\frac{Ne^2}{\varepsilon_0 m^*}} = \frac{2\pi c}{\lambda_p}, \quad \gamma = \frac{Ne^2 \rho}{m^*},$$

$$\varepsilon_{intrinsic}(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega), \quad (8)$$

ρ is the resistivity of silicon and N corresponds to the doping level. ω_p is the plasma frequency (λ_p is the plasma wavelength), γ is the relaxation ratio. Numerical plasmon frequency and relaxation values ω_p and γ are given in Table 1 from [16] considering a P -doping. $m^* = 0.34m_e$ is the effective mass of the charge carrier in the silicon crystal and m_e is the electron mass.

From Edwards' optical tabulated data (refractive index and absorption coefficient of silicon) on a large frequency range [15], we evaluate $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$. Eq. (8) gives a good representation of the dielectric function if the silicon is not degenerated (doping level $< 10^{21} \text{ cm}^{-3}$). The graphic representations of $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$ are given on Fig. 2(a) with a close zoom on deep UV frequencies on Fig. 2(b). This last figure shows the contribution to the permittivity of deep electrons from intra band transitions (d-band).

We know that the Kramers–Kronig relation is respected for silicon material from tabulated optical data. We can thus define as following the permittivity over the complex plan (see Eq. (6))

Table 1

Plasmon frequencies and relaxation ratios of silicon at different doping levels

Doping levels	$\omega_p (\times 10^{12} \text{ rad s}^{-1})$	$\gamma (\times 10^{13} \text{ s}^{-1})$	$\rho (\Omega \text{ cm})$
$1.1 \times 10^{15} \text{ cm}^{-3}$	3.20	1.182	13
$1.3 \times 10^{18} \text{ cm}^{-3}$	110.18	3.760	3.5×10^{-2}
$1.4 \times 10^{19} \text{ cm}^{-3}$	361.59	7.688	6.8×10^{-3}
10^{20} cm^{-3}	966.38	9.918	1.2×10^{-3}

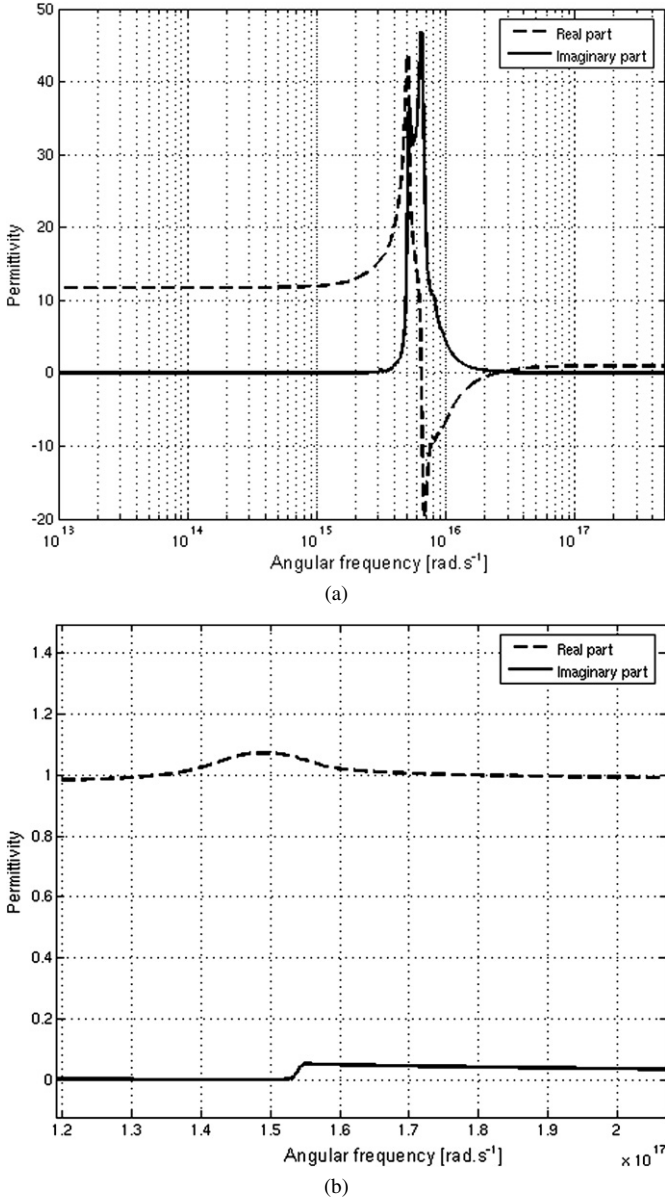


Fig. 2. (a) Real and imaginary parts of the intrinsic permittivity. (b) Zoom on the deep UV frequencies.

in order to implement Eq. (7):

$$\varepsilon(i\xi) = 1 + \frac{2}{\pi} \int_0^{+\infty} \frac{x\varepsilon_2(x)}{x^2 + \xi^2} dx. \quad (9)$$

From (9), we are able to plot the dielectric function $\varepsilon(i\xi)$, as shown in Fig. 3.

We note that this function tends to the static permittivity $\varepsilon_{Si} = 11.7$ (when the angular frequency tends to zero) and to unity value for very high frequency. With this model, we are able to compute properly the corrective factor for silicon slabs at different thicknesses.

To include the doping levels, two equivalent ways are available. We can first evaluate the Kramers–Kronig transform of the imaginary part of the complete dielectric function $\varepsilon(\omega)$ (8). We can also add the Kramers–Kronig transform (9) of the imagi-

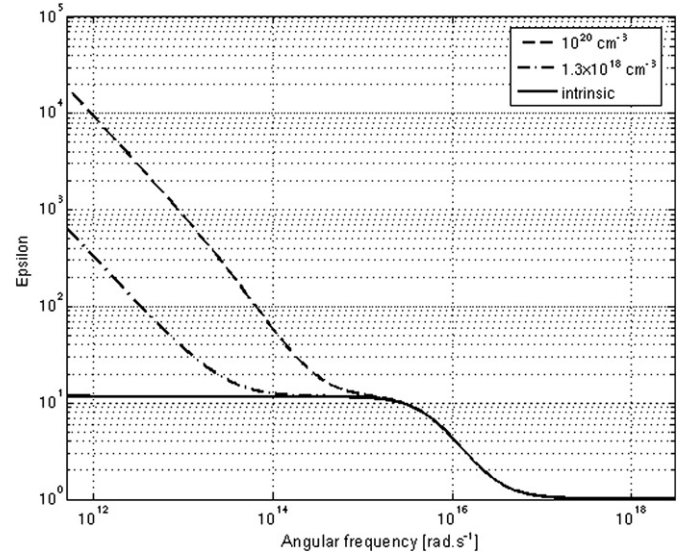


Fig. 3. Dielectric function $\varepsilon(i\xi)$ versus angular frequency from optical tabulated data for intrinsic silicon and two doping levels.

nary part of the intrinsic dielectric function $\varepsilon_2(\omega)$ to the imaginary part of the Drude expression. Both methods are equivalent because Drude model is causal.

For the considered gaps (above a few ten nanometers), we numerically verified that the contribution of angular frequencies lower than 10¹² rad.s⁻¹ was not significant for computations with about 1% accuracy. We plotted corrective factor at different thicknesses and doping levels for gap ranges of typical NEMS in Figs. 4 to 5.

For thick slabs, in Fig. 4(a), we note that the doping level strongly impacts the corrective factor. A quick look at Fig. 3 shows the contribution of free charge carriers on the shapes of complex permittivity due to the doping levels. We mention that this contribution is only at low frequency for realistic doping levels (below 10²¹ cm⁻³).

For small gaps, only the high frequencies of quantum fluctuation contribute strongly to the Casimir effect. The lower frequencies will become more and more influent when the gap will increase. Since the effect of doping on the dielectric function $\varepsilon(i\xi)$ happens only at low frequency, its influence will be visible essentially for large distances.

We can also explain this impact through the surface plasmon–polariton (SPP) coupling [17,18]. SPP coming from doping levels adds to coupling of plasmon modes from UV frequencies transitions (cf. Fig. 2(a) and (b)). The amplitude and the frequency of SPP modes grow with the doping level.

Fig. 4(b) shows the superposition of two different effects: the doping level influence and the slab thickness impact. For a given slab thickness (e.g. 100 nm), we noticed a decrement of the corrective factor from a particular distance threshold. This limit depends on both slab width and doping level. This last effect is clearly conspicuous on Fig. 4(b).

The slab thickness effect can be explained by comparing the width with the optical skin depth of the material δ [19] (cf. Eq. (10)), which is plotted on Fig. 6 for intrinsic, high doped

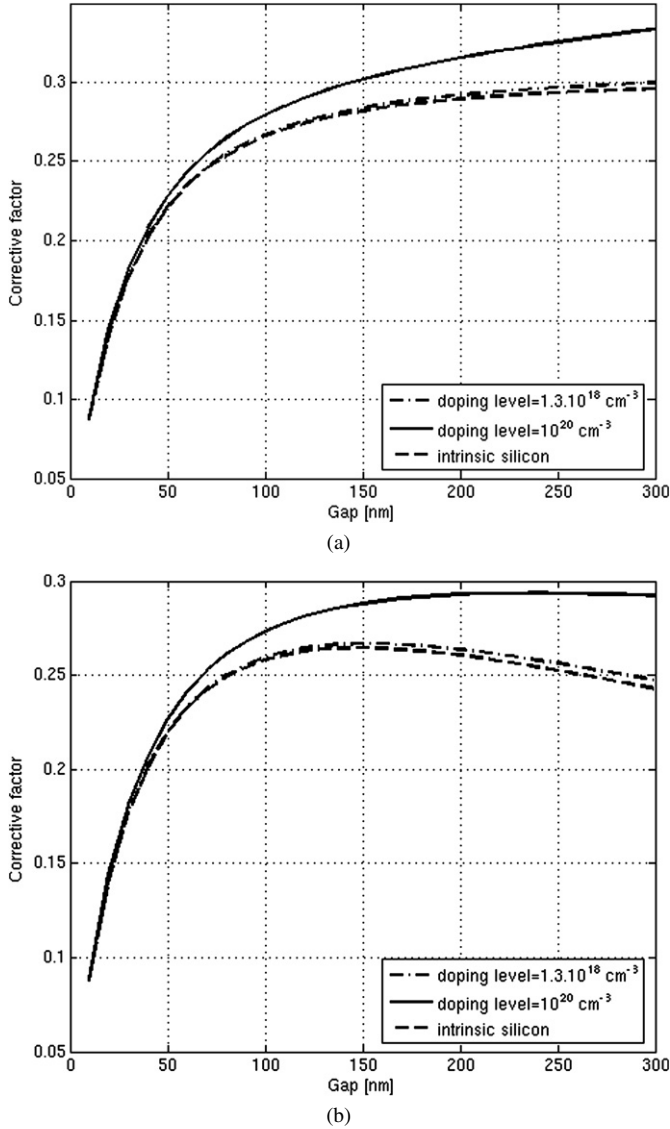


Fig. 4. (a) Corrective factor for thick silicon slabs (1 mm) versus gap at different doping levels. (b) Corrective factor for thin silicon slabs (100 nm) versus gap at different doping levels.

silicon and gold

$$\delta = \frac{c}{\omega k_a}, \quad (10)$$

ω is the angular frequency, c is the light velocity and k_a is the absorption coefficient of the material given by the tabulated optical data.

If the thickness is much larger than the skin depth, the slab can be considered as semi-infinite.

If we consider thin silicon slabs (typically 100 nm), by looking at the skin depth function (cf. Fig. 6), we notice that the slabs become transparent at angular frequencies lower than some $5 \times 10^{15} \text{ rad s}^{-1}$ and higher than $5 \times 10^{16} \text{ rad s}^{-1}$. In other words, the cavity remains really selective for a frequency range between 5×10^{15} and $5 \times 10^{16} \text{ rad s}^{-1}$.

For a given gap d , the characteristic fundamental mode of quantum fluctuations which contribute principally to the Casimir force is $\frac{c}{2d}$.

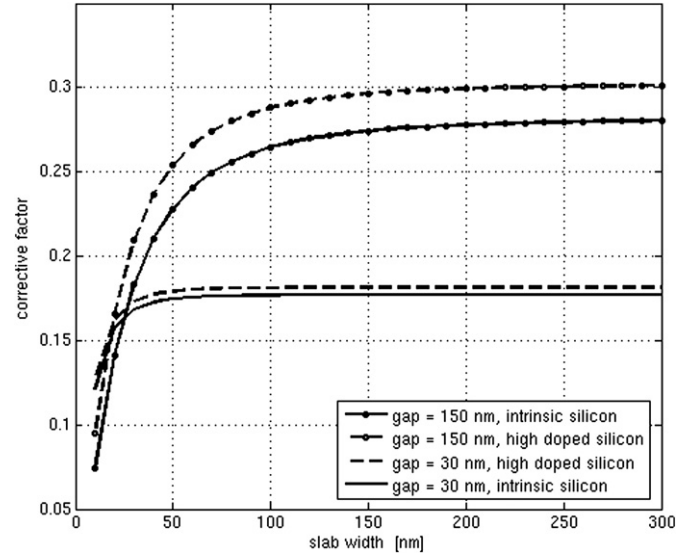


Fig. 5. Corrective factor for high doped silicon slabs versus slab thickness at different gaps.

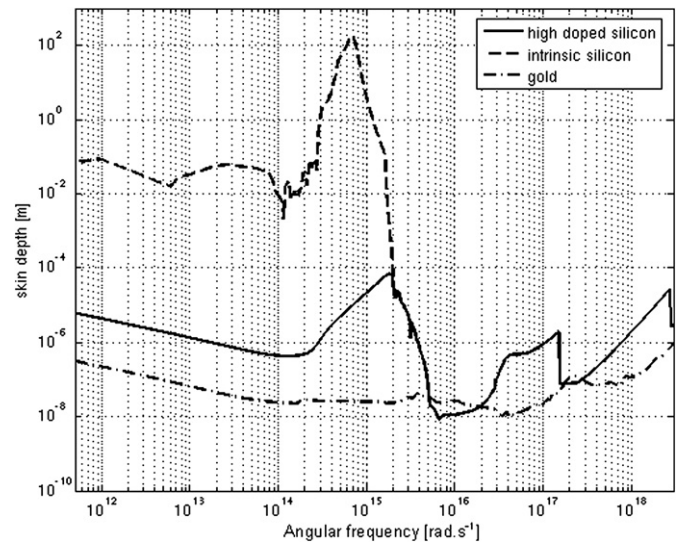


Fig. 6. Skin depth of intrinsic and high doped silicon compared to gold one.

Thus, for large gaps, the low frequencies (in the near IR band) impact the Casimir force. However, the cavity formed by thin slabs is strongly transparent at these frequencies. The more the gaps increase, the more the difference between thick and thin mirrors for the Casimir force will grow. Since the corrective factor for thick slab tends to the unity, the corrective factor for thin slab will thus decrease from a particular gap.

For small gaps, only high frequencies contribute to the Casimir force in case of perfect mirrors. Although the cavity is transparent at low frequencies for thin slabs (for example 100 nm), the energy carried by these frequencies is weak and its impact is unperceivable compared to the high frequency contributions. For the small gaps studied (between 10 nm and 100 nm), a cavity with thin slabs remains relatively similar to a cavity built with thick mirrors. The difference between both cases is relatively tiny (cf. Figs. 4 and 5). This is all

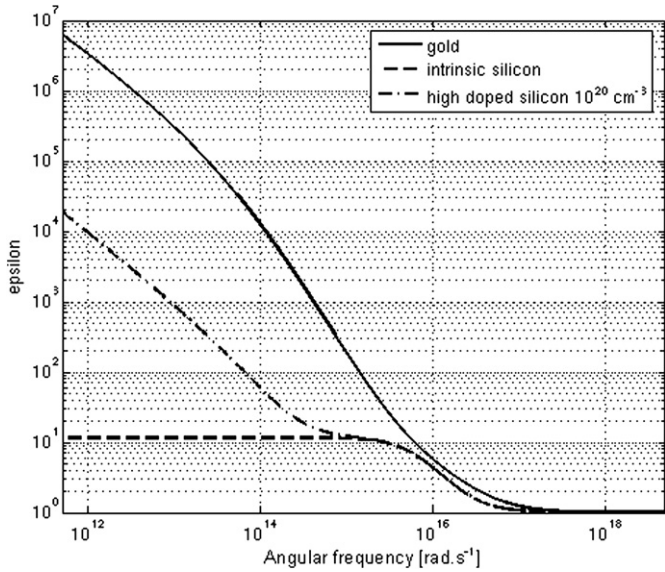


Fig. 7. Dielectric function $\varepsilon(i\xi)$ versus angular frequency computed from optical tabulated data for silicon and gold.

the more true as there are not relevant tabulated data beyond a few $10^{18} \text{ rad s}^{-1}$ available.

Finally, from a particular distance threshold which depends on the slab thickness, the system passes from one mode to the other.

4. Comparison with gold

We follow the same way to compute the Casimir force for gold case. The range of optical tabulated data [15] is nevertheless smaller than for silicon material. To be accurate enough and to stay in the same conditions, we use the Drude formula (11) to widen the data range towards lower frequencies. This method has already been justified in [14]

$$\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega(\omega - i\gamma)}, \quad \omega_p = \sqrt{\frac{Ne^2}{\varepsilon_0 m^*}} = \frac{2\pi c}{\lambda_p}, \quad (11)$$

ω_p is the plasma angular frequency, λ_p is the plasma wavelength. γ is the relaxation ratio. m^* is the effective mass of the charge carrier in the crystal, ε_0 is the permittivity of the vacuum, and N is the free carrier density. We choose $\omega_p = 13.71 \times 10^{15} \text{ rad s}^{-1}$ and $\gamma = 4.05 \times 10^{13} \text{ s}^{-1}$.

The dielectric function $\varepsilon(i\xi)$ is shown in Fig. 7 for silicon (intrinsic and high doping level) and gold.

In Fig. 8, we have plotted the corrective factors for thick and thin slabs of gold and silicon respectively for same gap range. We get similar results for thick gold mirrors as presented in Ref. [14]. However, we observe that the Casimir force is thus much more sensitive to slab thickness for silicon than for gold metal. This is perfectly understandable with Fig. 6. Whatever the thickness of the slab, we will nevertheless preserve good cavity selectivity for wide frequency range. The optical skin depth is perfectly scalable to plasma wavelength which is about 137 nm following a Drude model. High doped silicon behavior becomes similar to metals. Indeed, the dielectric function

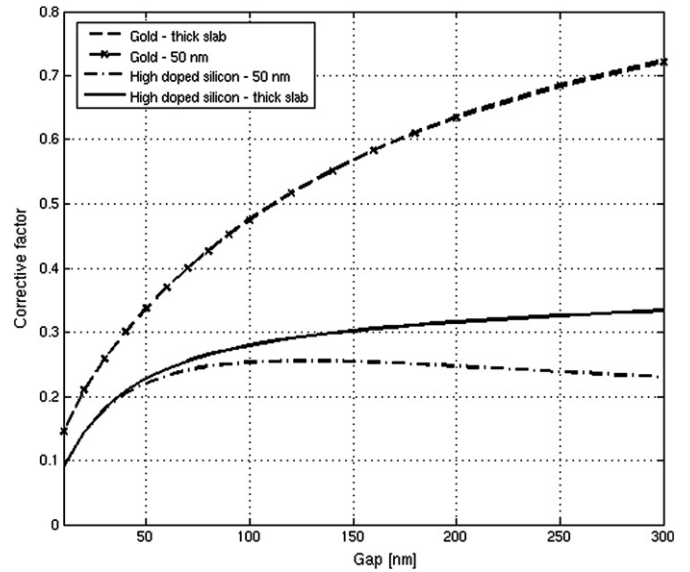


Fig. 8. Corrective factor for silicon and gold slabs versus gap at different doping levels.

$\varepsilon(i\xi)$ of silicon gets closer to gold when the doping level rises, especially for low frequencies range.

5. Summarize and discussion

Our study was focused on the Casimir force computation for silicon slabs and the results can be applied to any nanoelectromechanical systems for better static and dynamic modeling.

We have shown the specific behavior of this material compared to metals currently studied. We have underlined the slab thickness impact on the Casimir force for such material compared to metals like gold. This effect in addition to conductivity are especially of paramount in the silicon case. In particular, we have demonstrated a new behavior of the corrective factor which decreases from a particular gap threshold.

The results presented in this article clearly show that we have to include completely these parameters in the Casimir force computation for relevant nanostructures modeling. Of course, we clearly see that simple analytic models used for metals case (Drude or plasmon) cannot be applied to silicon whatever the considered gap. This justifies a complete computation from optical tabulated data if we need satisfying NEMS simulation models.

It might be also interesting to exploit this characteristic of silicon material displayed in this Letter for future experimental measurements of Casimir force.

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